SIMATS ENGINEERING

CO- 2: AT – 2: Simulation Task - Medium

COURSE NAME: 5G / 6G Communication Systems COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 2 - *Analyse LTE and 5G wireless network architectures, including carrier aggregation, MIMO techniques, beamforming strategies, and service-based core network functional entities.* (BL4)

Assessment Tool number : CO2-AT2
Assessment Tool Weightage : 25%
Assessment Tool Name :Simulation Task
Duration of this assessment : -
Marks : 20
Type of Assessment conduction: Home Assessment

Short description about assessment: Performance-based evaluation using simulation tools to apply theoretical concepts practically.

QUESTIONS: 1 Total Marks: 100

1. Simulation of LTE throughput performance under varying bandwidth, SNR, modulation order, coding rate, and user load (BL4)
2. Simulation of LTE protocol stack delay analysis under uplink traffic, downlink traffic, packet size, retransmission count, and channel quality (BL4)
3. Simulation of carrier aggregation performance in LTE-Advanced under single carrier, dual carrier, triple carrier, varying bandwidth, and SNR (BL4)
4. Simulation of LTE resource block allocation under different user numbers, scheduling methods, bandwidths, traffic load, and interference levels (BL4)
5. Simulation of LTE scheduler performance using Round Robin, Proportional Fair, Max C/I, delay variation, and throughput comparison (BL4)
6. Simulation of BER performance in 2×2, 4×4, 8×8 MIMO under varying SNR, modulation schemes, fading conditions, and antenna correlation (BL4)
7. Simulation of MIMO channel capacity under antenna number, SNR, bandwidth, fading model, and multiplexing order (BL4)
8. Simulation of SU-MIMO performance under channel gain, BER, throughput, latency, and SNR (BL4)
9. Simulation of MU-MIMO throughput comparison under user count, antenna count, SNR, modulation, and interference (BL4)

10. Simulation of higher-order MIMO diversity gain under fading channel, BER, outage probability, SNR, and antenna configuration (BL4)
11. Simulation of beamforming gain under steering angle, antenna elements, frequency, SNR, and interference (BL4)
12. Simulation of analog beamforming performance under phase shift variation, beam angle, antenna count, gain, and sidelobe level (BL4)
13. Simulation of digital beamforming under user direction, beam width, SINR, antenna spacing, and channel condition (BL4)
14. Simulation of hybrid beamforming under RF chains, beam direction, power allocation, SNR, and throughput (BL4)
15. Simulation of beam steering in antenna arrays under scan angle, frequency, gain, beam width, and sidelobe suppression (BL4)
16. Simulation of 5G throughput performance under QPSK, 16-QAM, 64-QAM, 256-QAM, and varying SNR (BL4)
17. Simulation of 5G NR air interface performance under different numerologies, bandwidths, latency, subcarrier spacing, and throughput (BL4)
18. Simulation of subcarrier spacing effect under 15 kHz, 30 kHz, 60 kHz, 120 kHz, and symbol duration (BL4)
19. Simulation of OFDMA resource allocation under user density, subcarriers, power allocation, throughput, and delay (BL4)
20. Simulation of waveform comparison between LTE OFDM and 5G NR under PAPR, BER, bandwidth, latency, and spectral efficiency(BL4)
21. Simulation of gNodeB traffic handling under user density, latency, throughput, packet loss, and scheduling (BL4)
22. Simulation of functional split in 5G RAN under processing delay, fronthaul load, throughput, latency, and energy consumption (BL4)
23. Simulation of RRC procedure timing under handover delay, signaling load, connection setup time, throughput, and failure rate (BL4)
24. Simulation of service-based architecture performance under AMF load, SMF load, UPF throughput, latency, and session count (BL4)
25. Simulation of network slicing under bandwidth allocation, latency, user priority, throughput, and resource utilization (BL4)
26. Simulation of AMF, SMF, and UPF traffic load under session count, packet delay, throughput, CPU utilization, and latency (BL4)
27. Simulation of control plane and user plane separation under packet delay, signaling load, throughput, latency, and efficiency (BL4)
28. Simulation of 5G core traffic routing under user density, mobility, latency, throughput, and resource utilization (BL4)
29. Simulation of packet delivery performance in 5G core under load variation, routing delay, throughput, packet loss, and queue size (BL4)
30. Simulation of mobility management under handover count, delay, throughput, signaling load, and failure probability (BL4)
31. Simulation of propagation loss under 700 MHz, 3.5 GHz, 28 GHz, 38 GHz, and 60 GHz (BL4)
32. Simulation of mmWave path loss under distance, frequency, atmospheric absorption, blockage, and antenna gain (BL4)

33. Simulation of channel modeling for mmWave MIMO under LOS, NLOS, fading, delay spread, and SNR (BL4)
34. Simulation of interference management under user density, reuse factor, SNR, SINR, and throughput (BL4)
35. Simulation of fading channel effects under Rayleigh, Rician, Nakagami, BER, and outage probability(BL4)
36. Simulation of V2X communication under latency, vehicle speed, packet delivery ratio, throughput, and interference (BL4)
37. Simulation of NB-IoT performance under coverage, delay, power consumption, throughput, and packet loss (BL4)
38. Simulation of edge computing delay under local processing, cloud processing, latency, bandwidth, and task size (BL4)
39. Simulation of smart city IoT traffic under sensor count, delay, packet delivery, bandwidth, and throughput (BL4)
40. Simulation of AI-based wireless optimization under prediction accuracy, throughput, delay, resource allocation, and energy efficiency (BL4)

Code of Conduct:

*I **S.Aravind (19241569)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
S.Aravind

RUBRICS FOR CALCULATION:

Simulation Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Conceptual Understanding	20%	Demonstrates strong understanding of underlying concepts in the simulation	Shows good understanding with minor gaps	Partial understanding; some misconceptions	Lacks understanding of key concepts

Model Design	25%	Develops accurate, well-structured simulation/model independently	Develops correct model with minor errors	Model is incomplete or requires guidance	Unable to develop a proper model
Tool Usage	20%	Uses simulation tools effectively with correct setup and execution	Uses tools with minor errors or guidance	Limited ability to use tools properly	Unable to use simulation tools
Analysis of Results	20%	Provides clear, logical analysis and meaningful interpretation of results	Analysis is reasonable with minor gaps	Superficial analysis; limited interpretation	Unable to analyze or interpret results
Documentation	15%	Presents results clearly with proper documentation and structured reporting	Documentation is adequate with minor issues	Poorly organized or incomplete documentation	No proper documentation or presentation

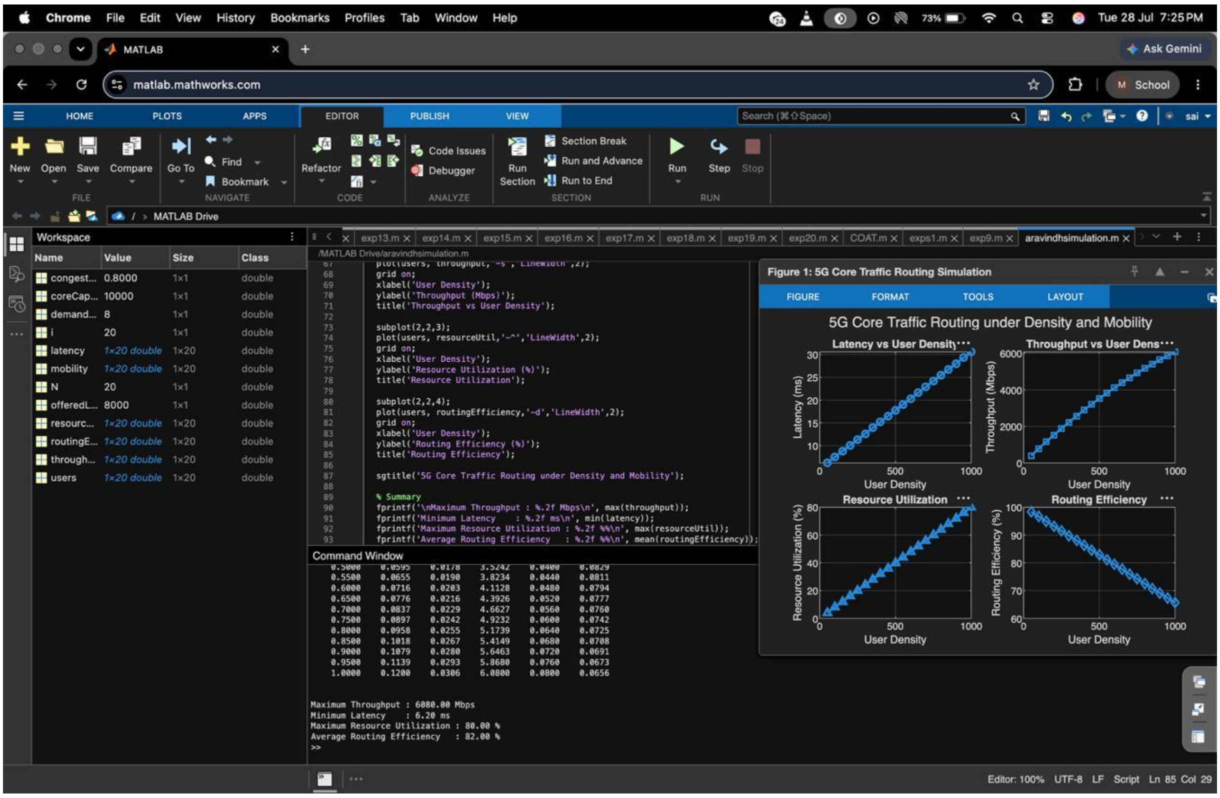
Code :

```
Clc;
Clear;
Close all;

% 5G Core Traffic Routing Simulation

% User density
Users = 50:50:1000;
N = length(users);

% Mobility speed (km/h)
Mobility = linspace(5,120,N);

% Core network capacity (Mbps)
coreCapacity = 10000;

% Demand per user (Mbps)
demandPerUser = 8;

% Arrays
```

```
Latency = zeros(1,N);
Throughput = zeros(1,N);
resourceUtil = zeros(1,N);
routingEfficiency = zeros(1,N);

for I = 1:N

   offeredLoad = users(i) * demandPerUser;

   % Resource utilization
   resourceUtil(i) = min(100, (offeredLoad/coreCapacity)*100);

   % Congestion factor
   Congestion = resourceUtil(i)/100;

   % Latency
   Latency(i) = 5 + 20*congestion + 0.08*mobility(i);

   % Throughput
   Throughput(i) = min(coreCapacity, offeredLoad) * …
               (1 – 0.002*mobility(i));

   If throughput(i) < 0
      Throughput(i) = 0;
   End

   % Routing efficiency
   routingEfficiency(i) = max(50, …
      100 – 0.25*resourceUtil(i) – 0.12*mobility(i));
End

% Display results
Disp('Users  Mobility(km/h)  Latency(ms)  Throughput(Mbps)  ResourceUtil(%)
RoutingEff(%)');
Disp([users' mobility' latency' throughput' resourceUtil' routingEfficiency']);

% Graphs
Figure('Name','5G Core Traffic Routing Simulation');

Subplot(2,2,1);
Plot(users, latency,'-o','LineWidth',2);
Grid on;
Xlabel('User Density');
Ylabel('Latency (ms)');
```

```
Title('Latency vs User Density');

Subplot(2,2,2);
Plot(users, throughput,'-s','LineWidth',2);
Grid on;
Xlabel('User Density');
Ylabel('Throughput (Mbps)');
Title('Throughput vs User Density');

Subplot(2,2,3);
Plot(users, resourceUtil,'-^','LineWidth',2);
Grid on;
Xlabel('User Density');
Ylabel('Resource Utilization (%)');
Title('Resource Utilization');

Subplot(2,2,4);
Plot(users, routingEfficiency,'-d','LineWidth',2);
Grid on;
Xlabel('User Density');
Ylabel('Routing Efficiency (%)');
Title('Routing Efficiency');

Sgtitle('5G Core Traffic Routing under Density and Mobility');

% Summary
Fprintf('\nMaximum Throughput : %.2f Mbps\n', max(throughput));
Fprintf('Minimum Latency    : %.2f ms\n', min(latency));
Fprintf('Maximum Resource Utilization : %.2f %%\n', max(resourceUtil));
Fprintf('Average Routing Efficiency   : %.2f %%\n', mean(routingEfficiency));
```

Simulation output :



MATLAB
matlab.mathworks.com
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
/MATLAB Drive/aravindhsimulation.m
Workspace
Name
Value
Size
Class
grid on;
xlabel('User Density');
ylabel('Throughput (Mbps)');
title('Throughput vs User Density');
subplot(2,2,3);
plot(users, resourceUtil,'-^','LineWidth',2);
grid on;
xlabel('User Density');
ylabel('Resource Utilization (%)');
title('Resource Utilization');
subplot(2,2,4);
plot(users, routingEfficiency,'-d','LineWidth',2);
grid on;
xlabel('User Density');
ylabel('Routing Efficiency (%)');
title('Routing Efficiency');
sgtitle('5G Core Traffic Routing under Density and Mobility');
% Summary
fprintf('\nMaximum Throughput : %.2f Mbps\n', max(throughput));
fprintf('Minimum Latency : %.2f ms\n', min(latency));
fprintf('Maximum Resource Utilization : %.2f %%\n', max(resourceUtil));
fprintf('Average Routing Efficiency : %.2f %%\n', mean(routingEfficiency));
Command Window
0.5500 0.0655 0.0190 3.8234 0.0440 0.0811
0.6000 0.0716 0.0203 4.1128 0.0480 0.0794
0.6500 0.0776 0.0216 4.3926 0.0520 0.0777
0.7000 0.0837 0.0229 4.6627 0.0560 0.0760
0.7500 0.0897 0.0242 4.9232 0.0600 0.0742
0.8000 0.0958 0.0255 5.1739 0.0640 0.0725
0.8500 0.1018 0.0267 5.4149 0.0680 0.0708
0.9000 0.1079 0.0280 5.6463 0.0720 0.0691
0.9500 0.1139 0.0293 5.8680 0.0760 0.0673
1.0000 0.1200 0.0306 6.0800 0.0800 0.0656
Maximum Throughput : 6080.00 Mbps
Minimum Latency : 6.20 ms
Maximum Resource Utilization : 80.00 %
Average Routing Efficiency : 82.00 %
Figure 1: 5G Core Traffic Routing Simulation
5G Core Traffic Routing under Density and Mobility
Latency vs User Density
Throughput vs User Density
Resource Utilization
Routing Efficiency
Latency (ms)
Throughput (Mbps)
Resource Utilization (%)
Routing Efficiency (%)
User Density